[METRIC]
H-B-391F
August 19, 1993
SUPERSEDING
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October 31, 1977

FEDERAL SPECIFICATION

BRUSHES, MOTTLING

The General Services Administration has authorized the use of this federal specification, for all federal agencies.

1. SCOPE AND CLASSIFICATION. This specification covers one type of mottling brush, in the following sizes.

SIZE mm.,	(inch)
13	(1/2)
25	(1)
38	(1-1/2)
64	(2-1/2)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-B-1880 -Brushes; Preparation for Delivery of
TT-T-291 -Thinner Paint, Mineral Spirits, Regular and Odorless

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Commercial Item Descriptions:

CID A-A-857 - Thinner, Dope and Laquer

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Specifications and Standards required by contractors in connection with specific acquisition functions should be obtained from the contracting officer.)

DISTRIBUTION STATEMENT: Approved for public release.
Distribution in unlimited

FSC 8020

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Federal Regulations:

29CFR 110.1200 -Hazard Communications Standard

2.2 **Other Publications.** The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Chemical Society (ACS) Specifications:

Reagent Chemicals (Latest Edition)
Hydrochloric Acid (HCl)

(Application for copies should be addressed to the American Chemical Society, 1155 Sixteenth St., N.W., Washington, D.C. 20036.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association Inc., Traffic Department, 2200 Mill Rd., Alexandria, VA 22314.)

American Society for Testing and Materials:

ASTM D13 - Specification for Spirits of Turpentine

(Application for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. **REQUIREMENTS**

3.1 **Materials and components.** Materials and components shall conform to the requirements specified herein.

3.1.1 **Hair.** The hair shall be soft, top quality, 100% pure, natural, undyed goat hair and shall be tested in accordance with 4.4.1. The test for coloring agent may be waived for hair that is certified by the supplier as being free of coloring agents.

3.1.2 **Setting Compound.** The setting compound shall be of such quality to withstand the test specified in 4.4.2.2, without softening or deteriorating, and without loosening of hairs in the setting compound.

3.1.3 **Ferrule.** The ferrule shall be made of corrosion resistant metal, nickel plated metal, or metal made corrosion resistant by a suitable coating. The coating shall not wrinkle, blister, or separate, nor become soft, dull or discolored when tested in accordance with 4.4.2.2. The ferrule depth shall be 32 mm (1-1/4 inch), minimum.

3.1.4 **Handle.** The handle shall be made of close grained hardwood, or a synthetic material with all sprues and runners removed. The synthetic handle shall not crack or break when tested as in 4.4.4 and shall show no visual change when tested in accordance with 4.4.2.2. Coated handles shall be tested in accordance with 4.4.2.1.

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3.2 **Construction and Physical Requirements**. The brushes shall conform to the physical requirements in Table I.

3.2.1 **Ferrule**. The ferrule shall be secured to the handle by nailing and a minimum of one nail on each side shall be used for the 13 mm (1/2 in), 25 mm (1 in.), and 38 mm (1-1/2 in) sizes. A minimum of two nails on each side shall be used for the 64 mm (2-1/2 in) size. The nails shall be 4.8 mm (3/16 in) long, minimum.

3.2.2 **Set**. The butt ends of the hair shall be set in a setting compound conforming to 3.1.2. The setting compound shall be at least 5 mm (3/16 in) above the bottom of the butt ends of the hairs.

3.2.3 **Handle**. The handle shall be a minimum of 127 mm (5 inches) long and shall be of mottling, pointy varnish, or tanalette style.

3.2.4 **Trim**. The trim shall be cupped or machined to a chisel edge with a minimum chisel of 6 mm (1/4 in). If cupped, flag ends shall be preserved. If machine chiseled, the ends shall be sanded to a fine, soft edge.

3.2.5 **Filler strips**. Filler strips shall be made of wood, metal, plastic, or cardboard, and shall not soften or otherwise change when tested as in 4.4.2.2.

Table I. Physical Requirements

SIZE	13 mm (1/2 in)	25 mm (1 in.)	38 mm (1-1/2 in)	64 mm (2-1/2)
FERRULE:				
Width, min., mm., (in)	13 (1/2)	25 (1)	38 (1-1/2)	64 (2-1/2)
Thickness, min., mm., (in)	6 (1/4)	8 (5/16)	8 (5/16)	10 (3/8)
HAIR:				
Length of hair outside of ferrule min., mm., (in)	21 (13/16)	24 (15/16)	27 (1-1/16)	32 (1-1/4)

3.3 **Marking for identification**. Unless otherwise specified (See 6.2), and in addition to any special marking required by the contract or order, each brush shall be stamped or otherwise permanently marked with the manufacturer's name or trade mark and the National Stock Number (NSN).

3.4 **Workmanship**. The finished brushes shall be clean, free of loose hair, well made, and free from any defects which may affect serviceability or appearance. There shall be no sharp edges, splinters, burrs, or other imperfections. The brushing part shall be trimmed evenly and uniformly to the finished length outside the ferrule.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection**. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements specified herein, using facilities acceptable to the

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Government. The Government reserves the right to perform any inspection herein, when deemed necessary to assure that the brushes conform to prescribed requirements.

4.2 **Inspection of preparation for delivery**. An examination shall be made to determine compliance with 3.2 and section 5 of this specification. The examination shall be conducted as specified in PPP-B-1880.

4.3 **Inspection of the end item**. Lots shall be assembled and randomly sampled in accordance with MIL-STD-105, inspection level I, with an acceptable quality level (AQL) of 2.5 percent defective. Sample brushes shall be examined for the following attributes:

Hair
Ferrules
Handles
Construction and assembly
Set
Trim
Workmanship
Identification markings

Any brush in the sample containing one or more visual or dimensional defect shall be considered a defective brush.

4.3.1 **Sampling for testing of the end item**. A random sample of brushes shall be selected from each lot in accordance with MIL-STD-105, inspection level S-2, with an acceptable quality level (AQL) of 2.5 percent defective.

4.4 **Testing procedure**. Samples selected in accordance with 4.3.1, shall be subjected to the following tests. Failure to pass any test shall be cause for rejection of the lot. All tests shall be evaluated for conformance to the requirements in section 3.

4.4.1 **Color test**. Samples of hair shall be soaked in boiling HCL (ACS Reagent Grade, 5 percent by volume in distilled water) for 5 minutes. The solution shall show no color other than green. Any other color shall be indicative of coloring matter.

4.4.2 **Solvent resistance.**

4.4.2.1 **Handle (Organic coating)**. Brush handles with organic coating shall be tested as follows:

A. With the brushing end up, approximately one-half of the handle shall be immersed, for 30 minutes, in a mineral spirits conforming to Federal Specification TT-T-291. After 30 minutes, the handle shall be removed from the mineral spirits and inspected for wrinkling, blistering, or separation of the coating.

B. With the brushing end up, approximately one-half of the handle shall be immersed, for 30 minutes, in turpentine conforming to ASTM D13. After 30 minutes, the handle shall be removed from the turpentine and inspected for wrinkling, blistering, or separation of the coating.

C. The brush handle shall be dried for 2 hours and the organic coating shall be examined for softening or dulling.

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4.4.2.2 **Base material, ferrule, filler strips, synthetic handles**. The brush shall be tested for solvent resistance using a thinner conforming to Commercial Item Description A-A-857.

(**NOTE**: The organic coating on the handle may be removed by this solvent. The objective is to test solvent resistance of the handle base material, plating/coating on the ferrule, filler strips, and the setting compound.)

The brush shall be suspended so that the brushing part and approximately one-half of the ferrule are immersed in the solvent for 1 hour. Synthetic handles shall be immersed to a depth of one-half of the handle length. The brush shall be removed and examined as follows:

A. **Handle and ferrule.** As soon as the brush is removed from the solvent, the organic coating on the ferrule shall be examined for wrinkling, blistering, or separation. The handle shall be examined for any deleterious effect by the solvent. The brush shall then be dried for 2 hours and examined for softening or dulling of the ferrule coating or deterioration of the handle base material.

B. **Filler strips.** When the brush is removed from the solvent, filler strips shall be examined for softening, swelling, or other deleterious change.

C. **Setting compound.** After removal of the brush from the solvent, the setting compound shall be examined for softening, loosening of bristle, and other solvent effects. After drying one hour, the brush shall be suspended by the handle, in a holding device, and a group of bristles, 6 mm (1/4 inch) in diameter, shall be given a tensile pull of 4.5 kg. (10 lbs) away from the ferrule, (type 1 brushes only). Repeat this tensile pull test, at random places, one time for each 25 mm (1 inch) of brush width. After thoroughly drying the brush, remove the ferrule and examine the setting compound for deterioration, bristle loosening, or other solvent damage.

4.4.3 **Drop test**. Remove hair and ferrule. Synthetic handles shall be dropped twice from a height of 3 m (10 feet) onto a concrete surface, and examined for compliance with 3.1.3 and 3.2.1.5.

PREPARATION FOR DELIVERY

5.1 **Preservation, packaging, packing and marking**. The preservation, packaging, packing, and marking shall be in accordance with PPP-B-1880. The level of packaging and packing shall be A,B,C, or otherwise specified in the contract or order (see 6.1). The marking shall be civil, military, or as specified (see 6.1).

6. NOTES

6.1 **Ordering data**. Purchasers should select the preferred options permitted herein, and include the following information in procure-ment documents:

(a) Title, number, and date of this specification.
(b) Type, grade, and size required (see 1).
(c) Levels of packaging and packing required, (see 5.1).
(d) Marking required (see 5.1).

PREPARING ACTIVITY:

GSA - FSS

CIVIL AGENCIES COORDINATING ACTIVITY:

GSA - FSS

JUS - FPI

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